

MOULIDHARAN S

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PROFILE

I am a motivated individual with a strong interest in Generative AI and machine learning, skilled in LLMs, Adaptive RAG, and multimodal AI systems, and passionate about building intelligent, scalable AI solutions while continuously learning emerging technologies.

INTERNSHIP EXPERIENCE - [GEN AI INTERN] INTROSPECT LABS LLP JUN 2025 – JAN 2026

AI-POWERED MEMORY PRESERVATION & INTERACTIVE DIGITAL TWIN

- Tools: Python, LLMs, Adaptive RAG, Qdrant, PostgreSQL, Neo4j, FastAPI, Sentence Transformers, Image Generation, Multimodal AI.
- Engineered a multimodal AI assistant using **LLMs** and **Adaptive Retrieval-Augmented Generation (RAG)** to deliver context-aware and intelligent responses.
- Integrated **Qdrant** (vector search), **PostgreSQL** (structured data), and **Neo4j** (graph-based reasoning) to enable efficient retrieval and enhanced contextual understanding.
- Built a multi-source knowledge system supporting text, image, audio, and video data, with **automated ingestion, chunking, and embedding generation**.
- Optimized AI pipelines using prompt engineering, embedding tuning, and efficient retrieval strategies to improve response accuracy, reduce latency, and enhance overall system performance.

PROJECTS

MAY-2026

- ECHOTALES – GENERATIVE AI STORYTELLING PLATFORM
- Tools: Python, Groq (Llama 3-70B), LangChain, Qdrant, Sentence Transformers, Whisper, gTTS, Diffusers (DreamShaper), MoviePy, Streamlit, Hugging Face, CUDA, Git
 - Built a multimodal AI storytelling platform that **Generates stories**, extracts **Character profiles**, and enables **Character-based conversations** using LLMs.
 - Integrated **AI Image Generation**, **Text-to-Speech (TTS)**, **Speech-to-Text (STT)**, and **Video Synthesis**, creating an end-to-end multimedia storytelling experience.
 - Implemented **Retrieval-Augmented Generation (RAG)** using **Qdrant** and **Sentence Transformers**, and developed an interactive Streamlit application leveraging Groq Llama 3-70B, LangChain, Whisper, Diffusers, and MoviePy for contextual story creation and narration.

RAGIFY – INTELLIGENT RETRIEVAL-AUGMENTED CHATBOT SYSTEM

APRIL-2025

- Tools: Python, LangChain, Qdrant, PostgreSQL, FAISS, Tavily API, Hugging Face, CUDA
- Engineered an AI-driven Retrieval-Augmented Generation (RAG) chatbot integrating Llama-4 (LLM), **Qdrant/PostgreSQL/FAISS** (Hybrid Vector Search), and **Tavily Search** (Real-Time Web Retrieval) to deliver accurate, context-aware, and scalable conversational intelligence.
- Built a multi-source knowledge ingestion pipeline enabling **document uploads**, automatic chunking, embedding generation, and metadata-aware semantic search for **domain-specific question answering**.
- Optimized retrieval with **hybrid similarity search**, embedding caching, reranking, and **asynchronous query execution** to minimize latency and maximize response precision.

EDUCATION

2021-2022

- Higher Secondary Certificate (HSC)
- Jothi Vidhyalaya Matriculation School - Percentage - 89%
- MSc in Decision and Computing Science
- Coimbatore Institute of Technology - CGPA - 8.20 (till 8thsemester)
- 2022 - Present

SKILLS

Programming: Python, HTML, CSS

Web Technologies: JavaScript, React (Basic), Angular (Basic), FastAPI

AI/ML & Generative AI: LLMs, Adaptive RAG, Prompt Engineering, Sentence Transformers, Multimodal AI, Image Generation

Databases: MySQL, MongoDB, PostgreSQL, Qdrant (Vector DB), Neo4j (Graph DB)

Tools & Platforms: Google Colab, MS Excel, Github, Postman

Soft Skills: Leadership, Teamwork, Critical Thinking

INTERESTS

Artificial Intelligence Machine Learning Open CV Data Visualization Deep Learning Natural Language Processing

EXTRA CURRICULAR

- CRICKET
- Current Vice Captain of Coimbatore Institute of Technology Cricket
 - Team Winners in CENTIES Inter College Cricket Tournamant in 2025
 - Runners in ZONALS Cricket Tournamant in 2025